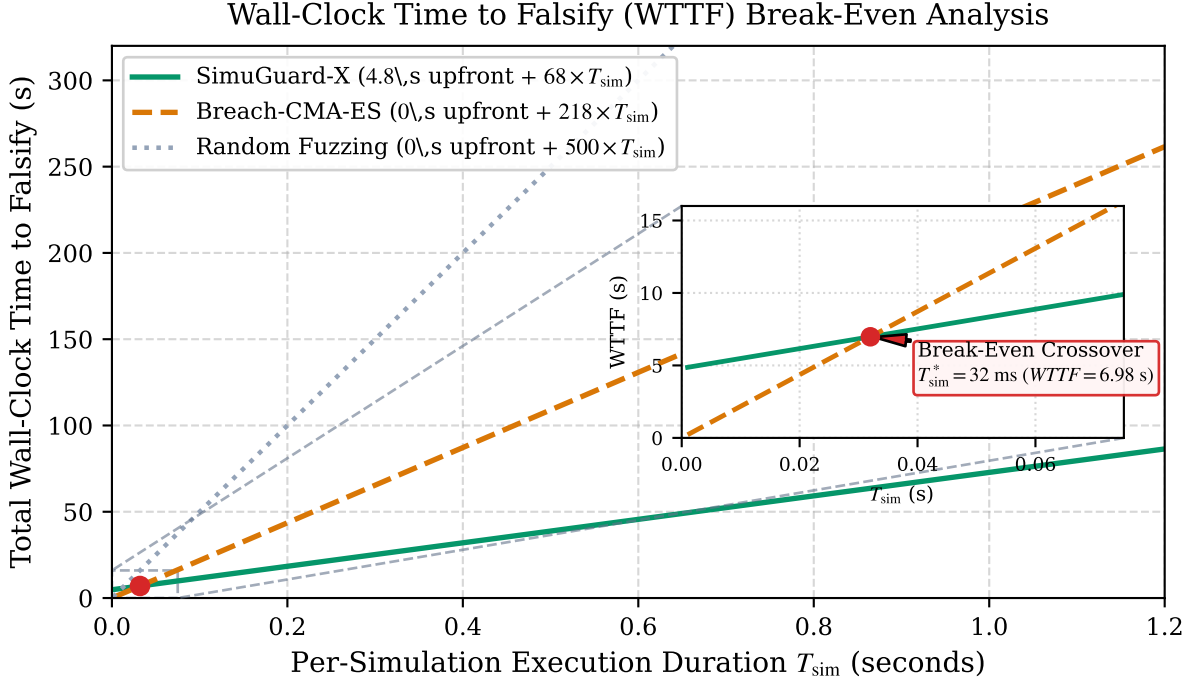
Wall-Clock Time to Falsify (WTTF) Break-Even Analysis
SimuGuard-X (4.8\,s upfront + 68×T_{sim})
Breach-CMA-ES (0\,s upfront + 218×T_{sim})
Random Fuzzing (0\,s upfront + 500×T_{sim})
Total Wall-Clock Time to Falsify (s)
Per-Simulation Execution Duration T_{sim} (seconds)
0
50
100
150
200
250
300
0.0
0.2
0.4
0.6
0.8
1.0
1.2
WTTF (s)
T_{sim} (s)
0
5
10
15
0.00
0.02
0.04
0.06
Break-Even Crossover
T^*_{sim} = 32 ms (WTTF = 6.98 s)